KILNDECK · DESIGN DECISION RECORD

KilnDeck concept selection

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DOCUMENT CONTROL

Document ID	KND-DDR-003	Version	0.1
Status	Draft	Date	2026-10-28
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/decisions/0003-concept-selection.md @ 96ed483		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-28	Amish Chadha	Concept selection between option A (rolling bridge deck) and option B (wall-top cableway); proposed, awaiting Amish

0003: Concept selection

- **Date:** 2026-10-28
- **Status:** proposed, awaiting Amish (KND-DEC-001 D10).

Context

Kit 1.8 (STANDARDS section 21) asks for at least two concept options with different working principles, scored against the user requirements. KilnDeck was designed before kit 1.8, so its current design, the rolling bridge deck taken to TRL 3 on 2026-10-03, is option A and is not redesigned. The STEM check (KND-STM-001) and both D3 models (`cad/src/model.py`, `cad/fidelity.json`) back the comparison in KND-OPT-001.

Options considered

1. **Option A, rolling bridge deck.** The fireman rides a guarded bridge over the fire on wall-top tracks and meters fuel with a rotor beside it. Weighted score 152 of 190.
2. **Option B, wall-top cableway.** The fireman stays on the outer wall top and sends a bucket out on a rope between two guyed masts, lowering and tipping it over each hole. Weighted score 124 of 190.

A third principle was considered and not modelled: long-handled tools or chutes worked from the wall tops. It cannot reach the middle rows of a 6 m trench (about 3 m from either wall), so it fails UR-003 on first-order reach alone.

Decision

Recommended: option A. It is 18 % ahead (outside the 10 % band that would send both to Amish without a pick), it protects against falls with a full guardrail where option B leaves the fireman at the trench edge of the wall top, it doses fuel rather than dumping a bucket, and it carries the complete TRL 3 record. Option B's advantages are cost (about USD 900 against USD 2,787, an estimate) and keeping the fireman off the fire's radiant heat. The runner-up idea worth grafting later: a small wall-top cableway for the curved kiln ends that option A does not serve.

Consequences

- If Amish confirms option A: `concept_chosen: A` in `project.yaml`; nothing else changes.
- If Amish chooses option B: the drawing, BOM, calculation note and build plan are redone for option B at TRL 3 (a new session).
- Either way the open items stay: the partner kiln survey (S9, S21), the team-handling question (D14) and the TRL 4 dose and floor temperature trials.